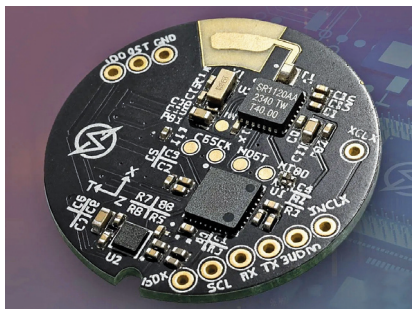




Presence detection kit

SPARK Microsystems has launched a presence detection kit (PDK) to help designers overcome limitations of legacy wireless platforms in smart

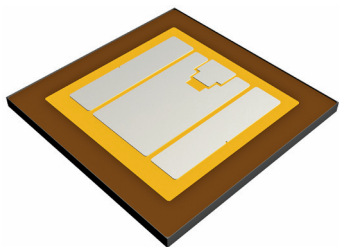


devices. Traditional UWB solutions also have high power use and cost. SPARK's LE-UWB technology addresses these issues. The PDK enables developers to implement LE-UWB, improving power consumption, battery life, and signal reliability while reducing development complexity and cost, thereby speeding time to market. Features include unidirectional and bidirectional communication, low-power beaconing with configurable detection zones, and line-of-sight time-of-flight for proximity and distance measurements. Compared with Bluetooth/BLE beaconing, LE-UWB uses over 10x less power; compared with UWB, it offers over 20x higher power efficiency.

SPARK Microsystems
<https://www.sparkmicro.com>

SiC power MOSFET

Wolfspeed's 10kV SiC power MOSFET is now commercially available, providing high-voltage performance for grid



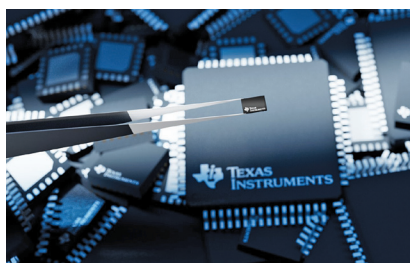
modernisation, industrial electrification, and AI data centres. The device can reduce system costs by about 30% by consolidating multi-cell designs and converting three-level inverters to two-level topologies. It increases power density by over 300% through higher switching frequencies, which

simplify gate/control circuits and reduce magnetics, while lowering thermal requirements by up to 50% and achieving 99% conversion efficiency compared with silicon IGBTs. With a rise time under 10ns, the MOSFET can replace mechanical spark-gap switches, eliminate arcing, improve energy transfer, and provide timing precision for applications including geothermal power, semiconductor plasma etching, sustainable fertiliser production, and AI data centres.

Wolfspeed
<https://www.wolfspeed.com>

Edge-AI microcontroller

Texas Instruments (TI) introduced two edge-AI microcontrollers: the MSPM0G5187 and AM13Ex, both featuring the TinyEngine neural pro-



cessing unit (NPU) for edge AI. The MSPM0G5187 runs neural network tasks alongside the main CPU, minimising flash use, cutting inference latency up to 90x, and reducing energy use over 120x, enabling AI on battery-powered devices. The AM13Ex combines an Arm Cortex-M33 core, TinyEngine NPU, and real-time motor control, supporting control and maintenance for up to four motors without extra components, lowering bill-of-materials costs by up to 30%, running AI for load sensing and energy optimisation, and including a trigonometric math accelerator that calculates 10x faster than CORDIC.

Texas Instruments (TI)
<https://www.ti.com>

Indoor climate sensor

Ligna Energy's Gwen, a battery-free indoor climate sensor reference design, harvests ambient energy, stores it in Ligna's S-Power 2S supercapacitor, measures temperature and humidity, and transmits data via Bluetooth Low Energy (BLE). By removing the need

for batteries, the sensor can operate maintenance-free and be placed flexibly in buildings. Ligna also demonstrates how ultra-thin energy storage

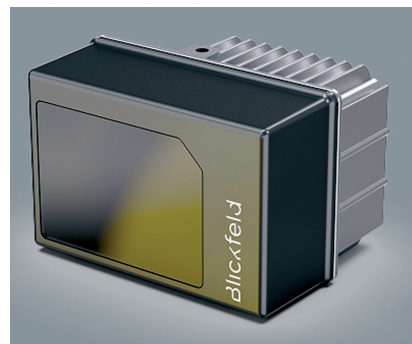


and minimal materials enable new form factors for indoor sensing and asset tracking. The focus is on providing adaptable design foundations for OEMs and solution providers, rather than a single sensor product, and on reducing maintenance and battery waste for long-term indoor operation.

Ligna Energy
<https://www.lignaenergy.com>

3D lidar sensor

Blickfeld's QbVolume is a 3D lidar sensor for real-time volume monitoring. It captures spatial data, processes it on-device, and provides automated measurements of volume, fill levels,



and material movement. It is used for raw material management, production, and logistics, tracking bulk materials, granulates, and pallets to support storage and transport across industries such as construction, mining, waste management, and distribution. QbVolume combines hardware, software, and a visualisation dashboard in a compact unit. Installation uses power-over-Ethernet or Wi-Fi, with brackets, cables, and weather covers for setup.

Blickfeld
<https://www.blickfeld.com>